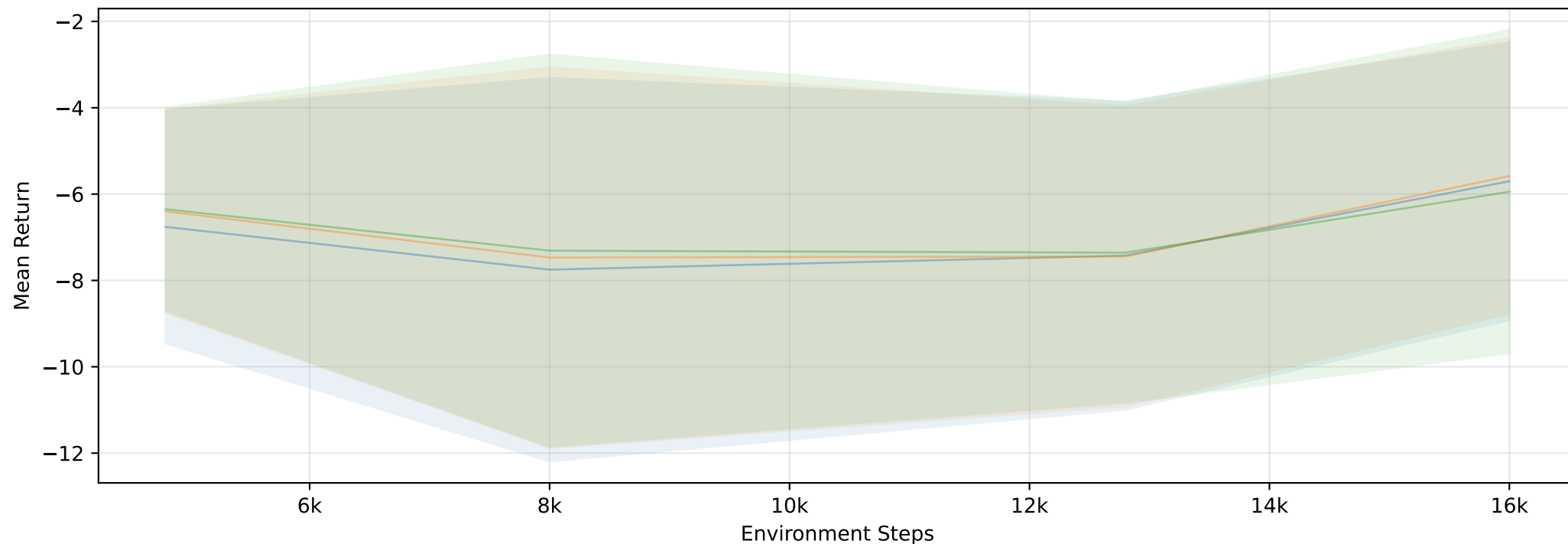
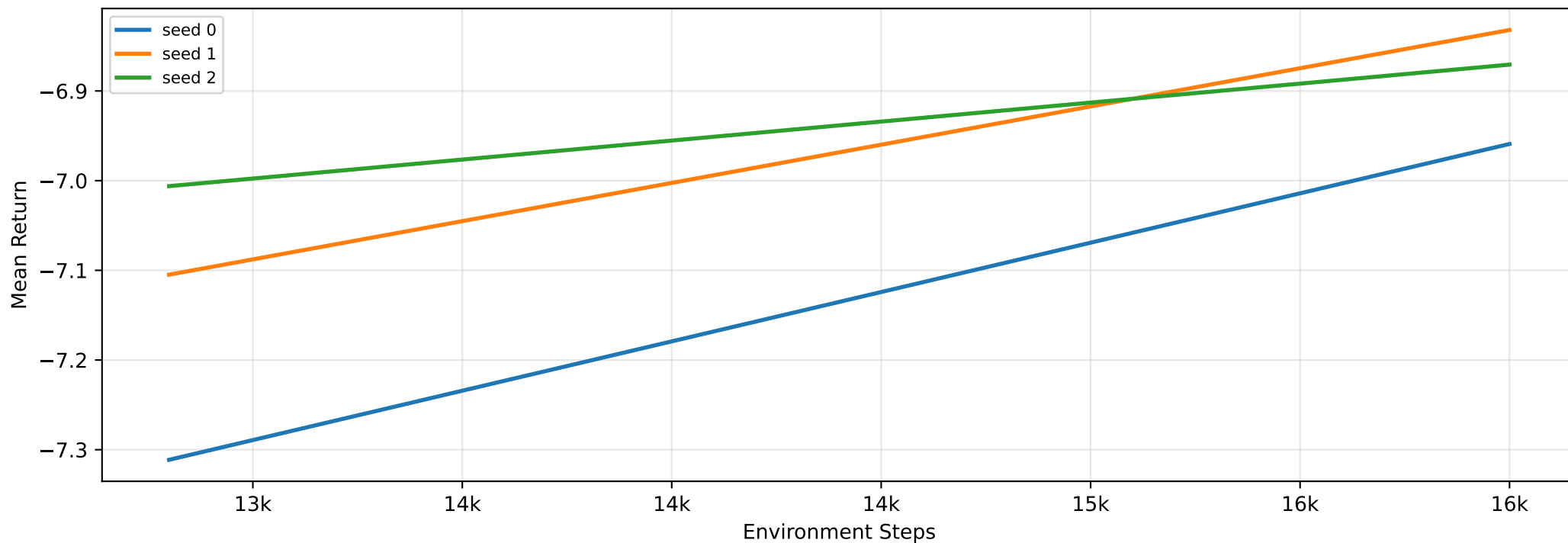


Structured Multi-Agent PPO — Evaluation Return

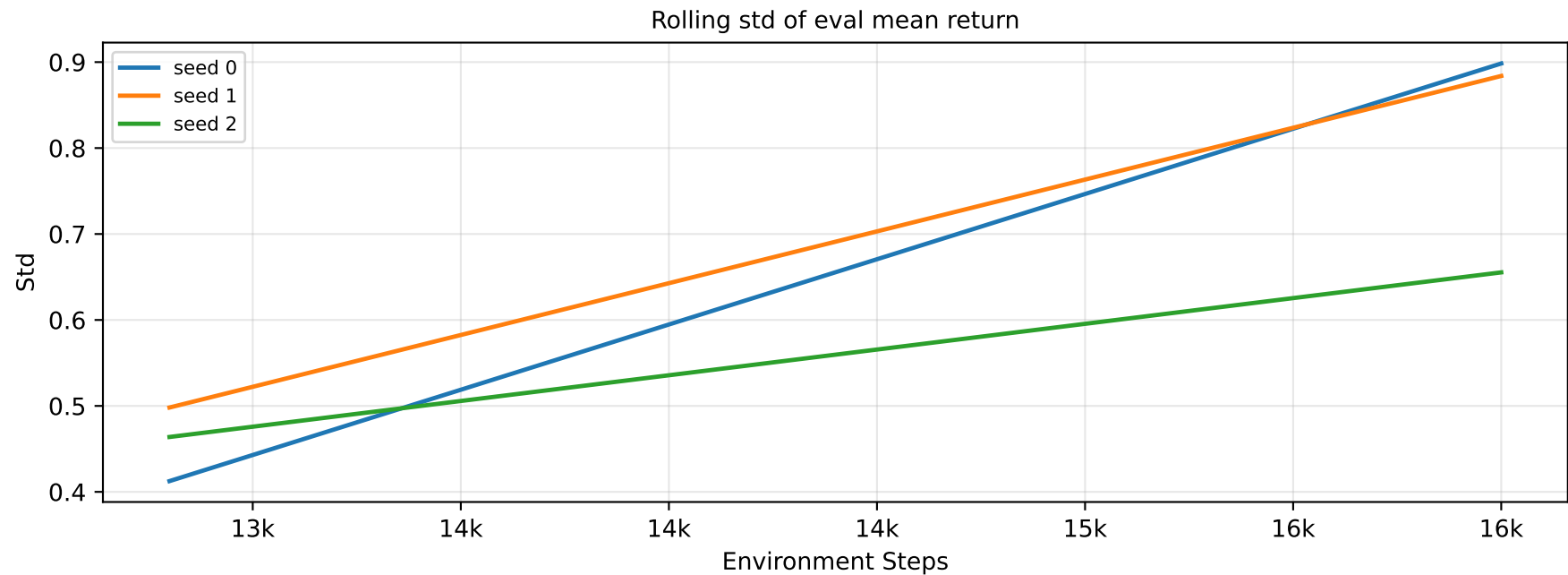
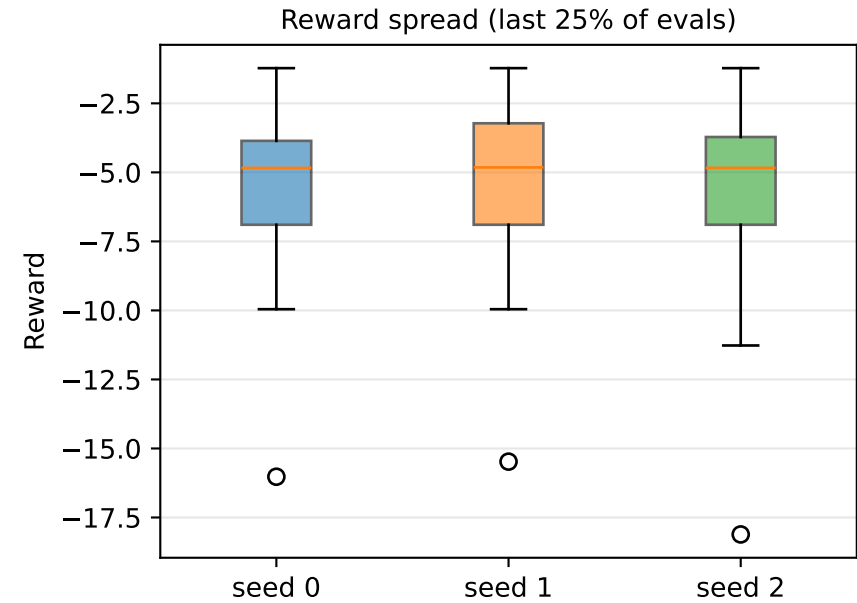
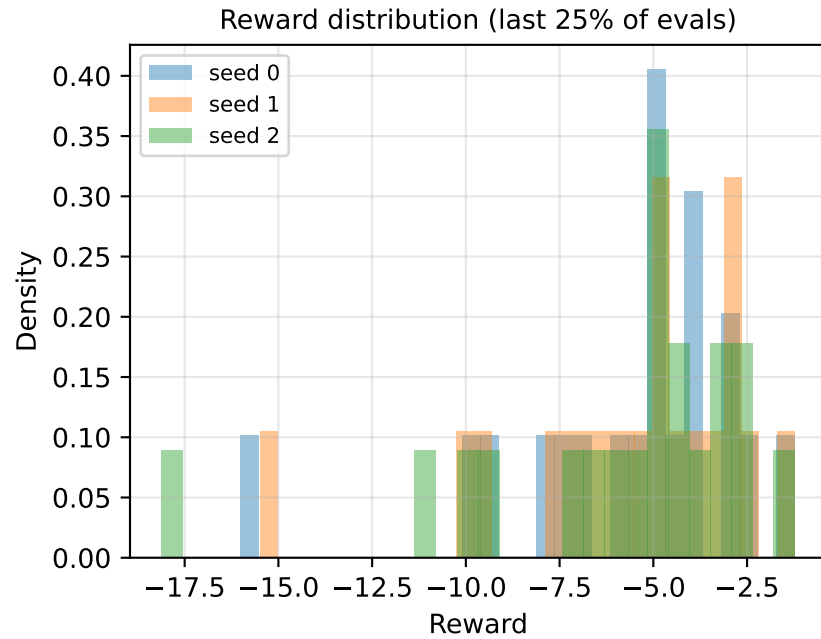
Raw evaluation mean return (± 1 std)



Smoothed evaluation mean return (window=3)



Structured Multi-Agent PPO — Distribution & Stability



Structured Multi-Agent PPO — Final Evaluation Summary

Run	Total steps	Mean return	Std	Ep length	Collisions	Landmarks covered	Success near end
seed 0	16k	-5.703	3.296	12.25	0.25	1.15	95.0%
seed 1	16k	-5.582	3.271	12.25	0.15	1.13	95.0%
seed 2	16k	-5.944	3.839	12.70	0.55	1.18	95.0%